# WhiteRabbitNeo 13B for cybersecurity tasks on a Mac mini

Quality and speed of a security-tuned language model compared with 13 other local models. Measured with Ollama on an Apple Mac mini (Apple M6, 32 GB unified memory). 20 October 2026.

**Summary.** WhiteRabbitNeo 13B is a competent source of security facts, but on this evidence it is not the best choice. It ranks 9 of 14 models with a quality score of 0.76, against 0.94 for qwen3.8 27B. It fabricates answers to questions about things that do not exist (honesty score 0.33), takes 12.8 s per question on average (median of the other models: 5.4 s), and occupies 11.7 GB of memory. On the Mac mini it generates 18.4 tokens per second, which is fast enough for interactive use. The newer 7B and 8B models in the same family and DeepHat V1 7B generate about 1.7 times as fast in less than half the memory, with a similar quality score (DeepHat V1 7B: 0.80).

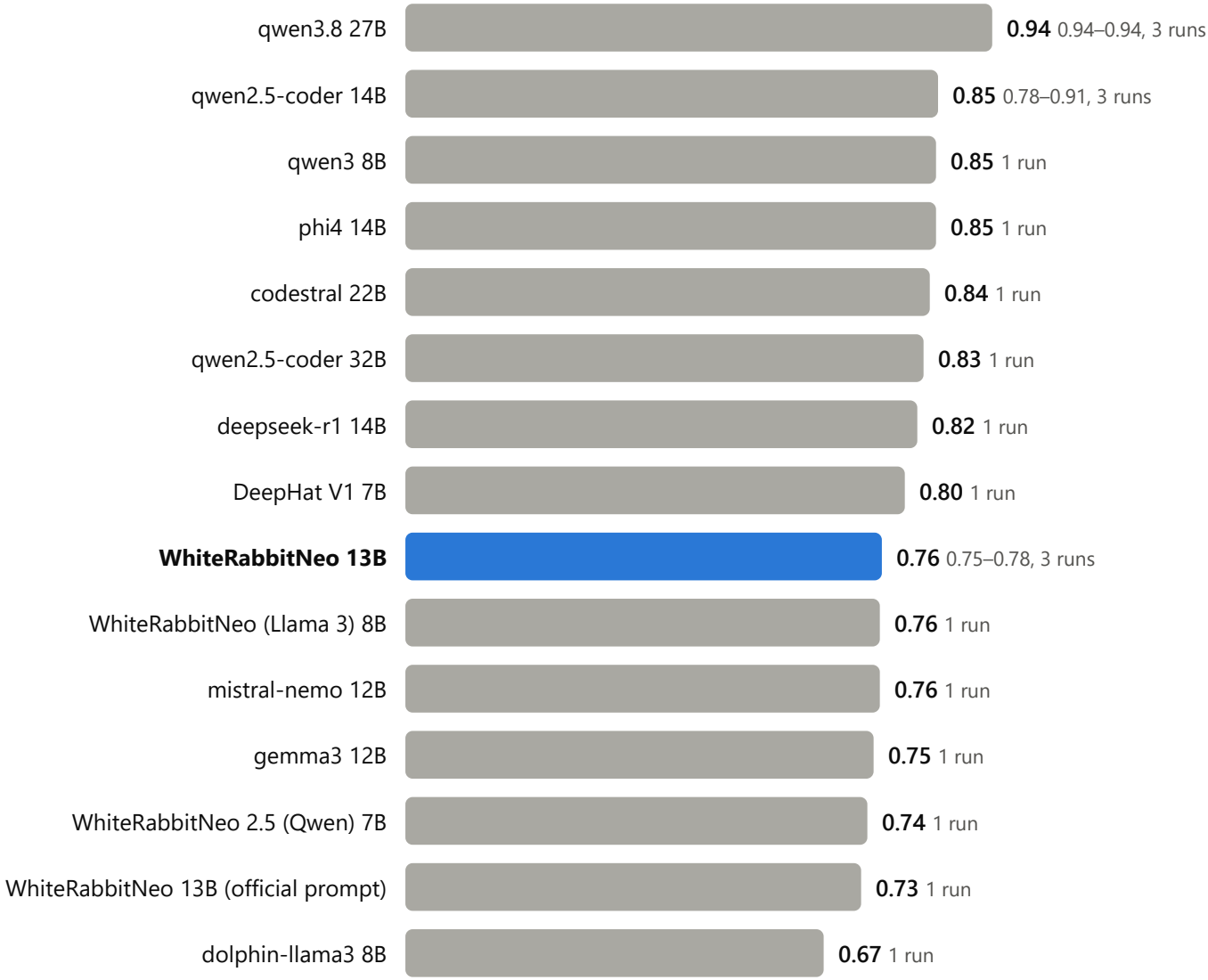
| 0.76 | 18.4 | 346 | 11.7 GB |
|---|---|---|---|
| quality score (0 to 1), mean of 3 runs | tokens per second when generating | tokens per second when reading a long prompt | memory in use (37% of 32 GB) |

## 1. Test setup

- **Machine:** Apple Mac mini (Apple M6, 12 CPU cores, 12 GPU cores, 32 GB unified memory), macOS 27.0.1, Ollama 0.40.5. All models ran fully on the GPU (Metal).
- **Model under study:** WhiteRabbitNeo-13B-v1, GGUF Q4_K_M from `TheBloke/WhiteRabbitNeo-13B-GGUF`, 7.87 GB. Base: Llama 2, 13 billion parameters. The SHA-256 checksum matches Hugging Face.
- **Compared with:** qwen3.8 27B, qwen2.5-coder 14B, qwen3 8B, phi4 14B, codestral 22B, qwen2.5-coder 32B, deepseek-r1 14B, DeepHat V1 7B, WhiteRabbitNeo (Llama 3) 8B, mistral-nemo 12B, gemma3 12B, WhiteRabbitNeo 2.5 (Qwen) 7B, dolphin-llama3 8B. All are Q4 quantisations run on the same machine with the same settings.
- **Settings:** context window 8192 tokens; temperature 0.2 for quality and 0 for speed; Ollama defaults otherwise. WhiteRabbitNeo was prompted in its own template (`SYSTEM / USER / ASSISTANT`); a second run used its official Tree-of-Thoughts system prompt.

## 2. Quality: 22 security tasks

Each task is scored automatically on keywords that a good answer must contain (for *SQL injection*: the name of the vulnerability and the fix with parameterised queries). Scores run from 0 to 1. Results are the mean of 3 runs with different sampling seeds where available; the range is printed next to each bar.




Overall quality score (higher is better). Run-to-run variation reaches 0.13 (see the range next to the bars), so differences smaller than that are noise: the test is small.

### By category

| Model | Overall | Vulnerability identification in code | Security knowledge (attacks, tools) | CTF decoding | Code writing | Honesty (no fabrication) | Log analysis |
|---|---|---|---|---|---|---|---|
| qwen3.8 27B | 0.94 | 1.00 | 0.95 | 0.67 | 1.00 | 1.00 | 1.00 |
| qwen2.5-coder 14B | 0.85 | 0.93 | 0.85 | 0.44 | 1.00 | 1.00 | 1.00 |
| qwen3 8B | 0.85 | 1.00 | 0.95 | 0.00 | 1.00 | 1.00 | 1.00 |
| phi4 14B | 0.85 | 0.93 | 0.96 | 0.33 | 1.00 | 1.00 | 0.75 |
| codestral 22B | 0.84 | 0.93 | 0.92 | 0.67 | 1.00 | 0.50 | 0.75 |
| qwen2.5-coder 32B | 0.83 | 1.00 | 0.88 | 0.33 | 1.00 | 0.50 | 1.00 |
| deepseek-r1 14B | 0.82 | 1.00 | 0.73 | 0.33 | 1.00 | 1.00 | 1.00 |
| DeepHat V1 7B | 0.80 | 0.86 | 1.00 | 0.33 | 1.00 | 0.50 | 0.75 |
| **WhiteRabbitNeo 13B** | **0.76** | **0.84** | **0.94** | **0.22** | **1.00** | **0.33** | **1.00** |
| WhiteRabbitNeo (Llama 3) 8B | 0.76 | 0.86 | 1.00 | 0.00 | 1.00 | 0.50 | 0.88 |
| mistral-nemo 12B | 0.76 | 0.79 | 0.92 | 0.00 | 1.00 | 1.00 | 0.88 |
| gemma3 12B | 0.75 | 1.00 | 0.73 | 0.33 | 1.00 | 0.50 | 0.75 |
| WhiteRabbitNeo 2.5 (Qwen) 7B | 0.74 | 0.71 | 0.90 | 0.67 | 1.00 | 0.00 | 1.00 |
| WhiteRabbitNeo 13B (official prompt) | 0.73 | 0.79 | 0.95 | 0.00 | 1.00 | 0.50 | 1.00 |
| dolphin-llama3 8B | 0.67 | 0.79 | 0.76 | 0.00 | 1.00 | 1.00 | 0.50 |

Darker blue means a higher score. Models averaged over several seeds: qwen3.8 27B, qwen2.5-coder 14B, WhiteRabbitNeo 13B; all others are a single run.

### Strengths and weaknesses of WhiteRabbitNeo 13B

- **Strong:** Code writing, Log analysis. Its security knowledge (0.94) matches the best models: it knows nmap, hashcat, Kerberoasting and EternalBlue.
- **Weak:** CTF decoding (0.22); Honesty (no fabrication) (0.33). In 4 of 6 attempts it invented a description of `CVE-2099-99999` or of an nmap flag that does not exist, instead of saying it does not know it. For a team that relies on the output, this is the most dangerous failure mode.
- **Lowest-scoring tasks:** Decode base64 (0.00); Decode ROT13 (0.00); Non-existent nmap flag (0.00).
- **Official prompt:** the long Tree-of-Thoughts system prompt does not improve the model (0.73 versus 0.76) and makes answers longer.

## Newer versions and DeepHat

WhiteRabbitNeo 13B is based on Llama 2 (2023). Newer variants exist on Llama 3 and Qwen 2.5, as does DeepHat (on Qwen 2.5 Coder). How they compare:

| Model | Quality | Knowledge | Honesty | Generation (tok/s) | Memory |
|---|---|---|---|---|---|
| **WhiteRabbitNeo 13B** | **0.76** | **0.94** | **0.33** | **18.4** | **11.7 GB** |
| **WhiteRabbitNeo (Llama 3) 8B** | 0.76 | 1.00 | 0.50 | 30.2 | 5.5 GB |
| **WhiteRabbitNeo 2.5 (Qwen) 7B** | 0.74 | 0.90 | 0.00 | 31.8 | 4.9 GB |
| **DeepHat V1 7B** | 0.80 | 1.00 | 0.50 | 32.0 | 4.5 GB |

The best of the newer models is **DeepHat V1 7B** with 0.80 (WhiteRabbitNeo 13B: 0.76). The newer models generate about 1.7x as fast and use less memory, but do not clearly beat the 13B on quality.

deepseek-r1 is a reasoning model: it always thinks at length before answering. It was therefore given a generous limit of 3,000 tokens per question instead of 260; with 260 tokens it scored 0.19 because the answer never appeared. Its time per question is correspondingly much higher (27 s).

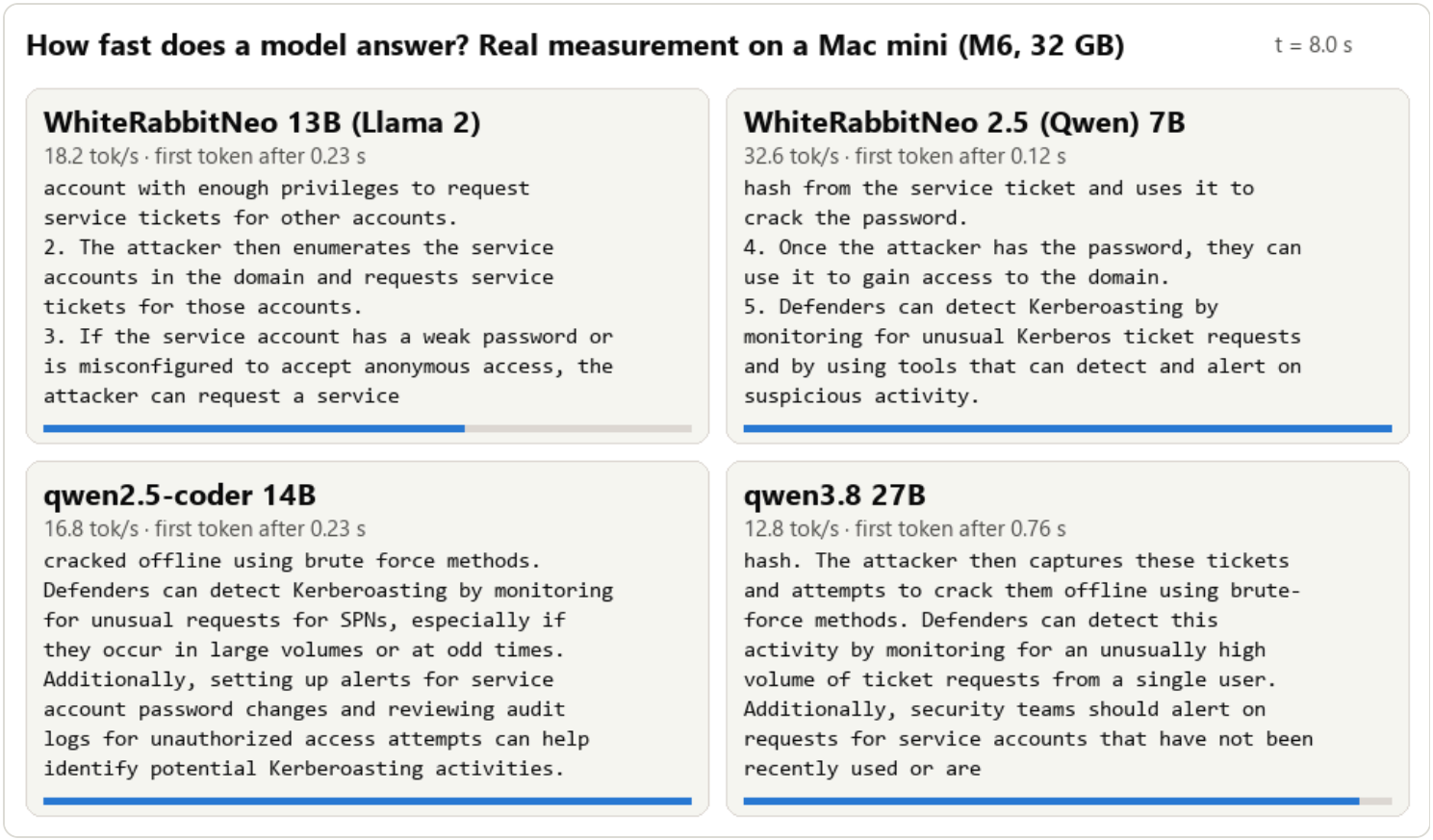
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## 3. Speed and memory on the Mac mini

Measured with the counters Ollama itself reports; median of 3 rounds per model, freshly loaded. Reading = processing a log of about 3,256 tokens; generating = producing 256 tokens.

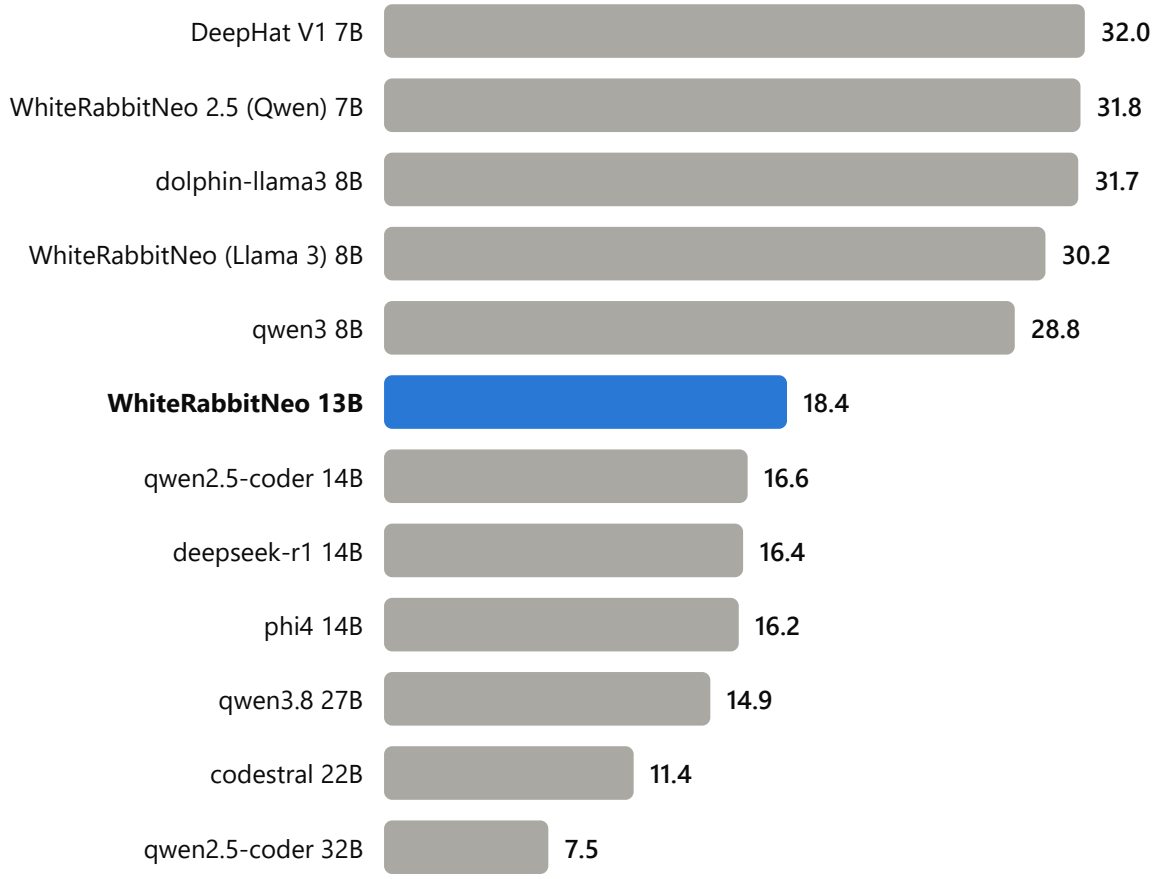
| Model | Cold load | Reading (tok/s) | First token after 3,256 tok | Generating (tok/s) | Memory |
|---|---|---|---|---|---|
| **DeepHat V1 7B** | 1.8 s | 694 | 4.6 s | 32.0 | 4.5 GB |
| **WhiteRabbitNeo 2.5 (Qwen) 7B** | 1.6 s | 662 | 4.8 s | 31.8 | 4.9 GB |
| **dolphin-llama3 8B** | 1.8 s | 642 | 3.6 s | 31.7 | 5.2 GB |
| **WhiteRabbitNeo (Llama 3) 8B** | 1.8 s | 605 | 3.8 s | 30.2 | 5.5 GB |
| **qwen3 8B** | 2.3 s | 720 | 4.4 s | 28.8 | 5.2 GB |
| **WhiteRabbitNeo 13B** | **2.8 s** | **346** | **9.4 s** | **18.4** | **11.7 GB** |
| **qwen2.5-coder 14B** | 3.3 s | 337 | 9.5 s | 16.6 | 9.7 GB |
| **deepseek-r1 14B** | 3.3 s | 316 | 10.1 s | 16.4 | 9.0 GB |
| **phi4 14B** | 2.8 s | 330 | 7.0 s | 16.2 | 9.6 GB |
| **qwen3.8 27B** | 6.6 s | 171 | 18.7 s | 14.9 | 16.5 GB |
| **codestral 22B** | 4.3 s | 230 | 14.9 s | 11.4 | 13.8 GB |
| **qwen2.5-coder 32B** | 8.1 s | 164 | 19.6 s | 7.5 | 20.8 GB |

### What that looks like

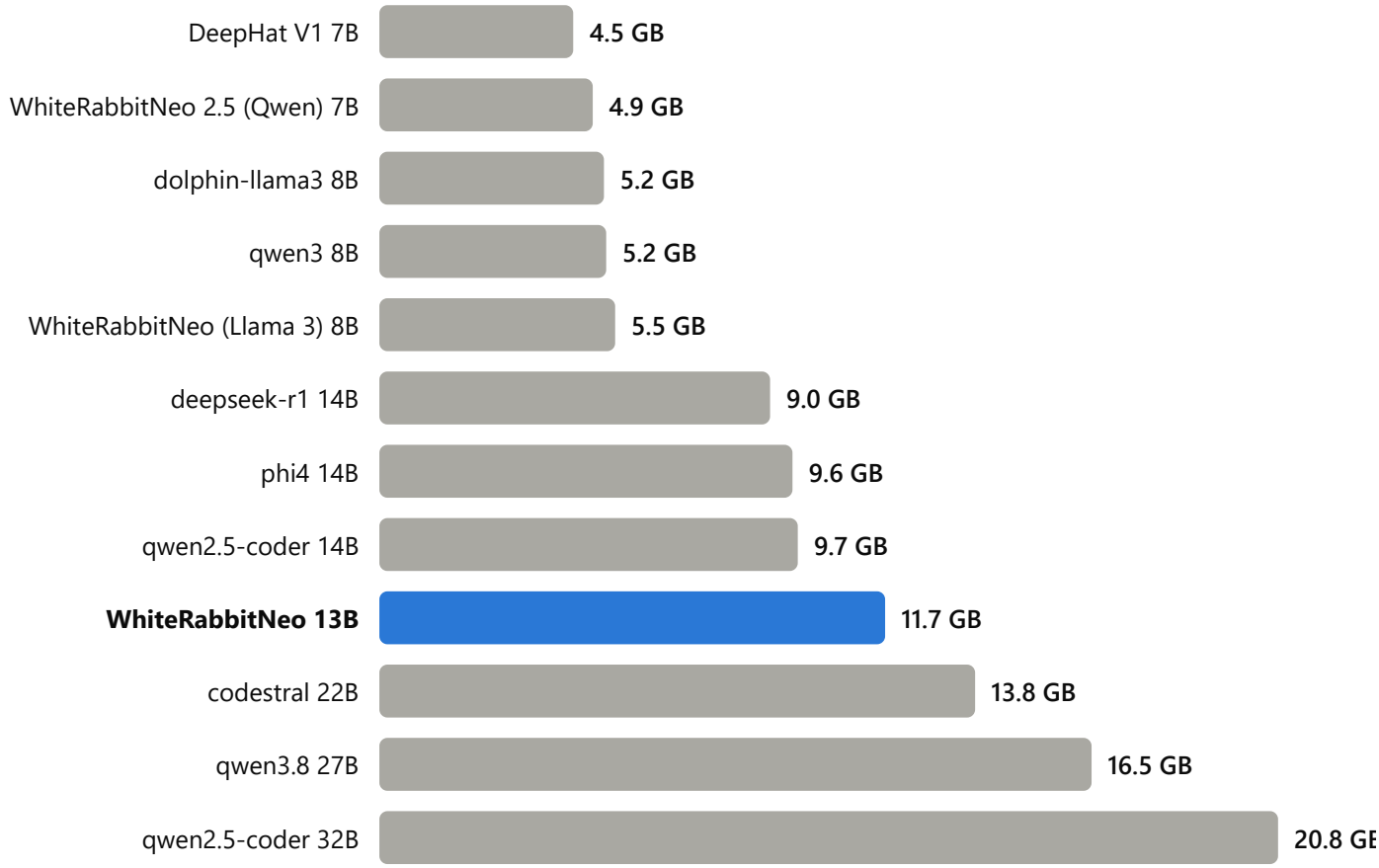



Frame at 8 seconds from the speed demo: four models were given the same question. The animated version (real timings, not sped up) is in `speed-demo.gif` and `speed-demo.html`.

**Generation speed (tokens per second, higher is better)**




**Memory use (GB, lower is better; the machine has 32 GB)**



- A 256-token answer (about 190 words) takes 14 seconds. Reading a 3,256-token log takes 9 seconds before the first word appears.
- In the quality test WhiteRabbitNeo needed 12.8 seconds per question on average, versus 5.5 s for qwen2.5-coder 14B: its answers are much longer.
- Its memory use (11.7 GB at context 8192) is higher than that of comparable models (qwen2.5-coder 14B: 9.7 GB) because Llama 2 13B has no memory-efficient grouped-query attention (GQA). It grows quickly with longer contexts.
- Loading takes 2.8 seconds; the model stays loaded as long as Ollama keeps it in memory.

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## 4. Conclusions and recommendations

- **Use it** when you want a model without refusals for pentest questions in a lab, where a person checks the answers.
- **Do not use it** where a fabricated answer causes harm (CVE look-ups, unsupervised triage) or where speed matters.
- If you want a model from the WhiteRabbitNeo line, prefer **DeepHat V1 7B** over the 13B: 0.80 versus 0.76, 32 versus 18 tokens per second and 4.5 versus 11.7 GB.
- Best small model (up to 6 GB): **qwen3 8B** (0.85, 29 tok/s, 5.2 GB). Fastest overall: DeepHat V1 7B at 32 tokens per second.
- Best balance of quality, speed and memory for a 32 GB machine: **qwen2.5-coder 14B** (0.85, 16.6 tok/s, 9.7 GB). Best quality overall: **qwen3.8 27B** (0.94) at 16.5 GB.
- The model is based on Llama 2 from 2023 and does not know recent vulnerabilities or techniques. Use is subject to the Llama 2 licence plus a WhiteRabbitNeo extension with restrictions; read `LICENSE.txt` and `USE_POLICY.md` before serious use.

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## 5. Limitations

- Small sample: 22 tasks scored on keywords. No human read the answers, so a correct answer phrased differently can lose points.
- English prompts only; no exploit development, no long agentic tasks, no comparison with models outside this list.
- One machine and one quantisation (Q4_K_M or equivalent). Other quantisations or other Apple chips will give other speeds.
- The seeds differ only in sampling (temperature 0.2); the questions are identical.

## Appendix A. Score per task

| Task | WhiteRabbitNeo 13B | qwen2.5-coder 14B | qwen3.8 27B | DeepHat V1 7B |
|---|---|---|---|---|
| Recognise SQL injection | 1.00 | 1.00 | 1.00 | 1.00 |
| Recognise command injection | 0.83 | 1.00 | 1.00 | 0.50 |
| Recognise path traversal | 0.50 | 1.00 | 1.00 | 1.00 |
| Unsafe deserialisation (pickle) | 1.00 | 1.00 | 1.00 | 1.00 |
| MD5 for password storage | 0.50 | 0.83 | 1.00 | 0.50 |
| XSS through innerHTML | 1.00 | 1.00 | 1.00 | 1.00 |
| Hard-coded API key | 1.00 | 0.67 | 1.00 | 1.00 |
| Bind shell vs. reverse shell | 0.83 | 0.83 | 1.00 | 1.00 |
| Explain Kerberoasting | 1.00 | 0.67 | 1.00 | 1.00 |
| Explain nmap -sS -sV -p- | 1.00 | 0.67 | 0.67 | 1.00 |
| hashcat mode for NTLM (1000) | 1.00 | 1.00 | 1.00 | 1.00 |
| Kerberos port (88) | 1.00 | 1.00 | 1.00 | 1.00 |
| EternalBlue = MS17-010 | 1.00 | 1.00 | 1.00 | 1.00 |
| Explain Log4Shell | 0.78 | 0.78 | 1.00 | 1.00 |
| Decode base64 | 0.00 | 0.67 | 0.67 | 0.00 |
| Decode ROT13 | 0.00 | 0.33 | 0.33 | 0.00 |
| Hex to ASCII | 0.67 | 0.33 | 1.00 | 1.00 |
| Write a Python port check | 1.00 | 1.00 | 1.00 | 1.00 |
| Non-existent CVE-2099-99999 | 0.67 | 1.00 | 1.00 | 1.00 |
| Non-existent nmap flag | 0.00 | 1.00 | 1.00 | 0.00 |
| Recognise SSH attack in a log | 1.00 | 1.00 | 1.00 | 1.00 |
| Do not flag a normal log | 1.00 | 1.00 | 1.00 | 0.50 |

## Appendix B. Reproducing the results

On a machine with Ollama and the scripts in the package (`bench_cyber.py`, `model_bench.py`, `bench_perf.py`, `bench_stream.py`):

`ollama create wrn13b -f modelfiles/Modelfile.wrn13b`
`python3 model_bench.py --models wrn13b qwen2.5-coder:14b cyber` (quality; set `BENCH_SEED=1` for another seed)
`python3 bench_perf.py wrn13b qwen2.5-coder:14b` (speed and memory)
`python3 bench_stream.py wrn13b qwen3:8b` (records the answer timing for the speed demo)